

Tasks Us - Hands On Workshop

Microsoft Philippines

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Fabric Data Agent

What is Fabric Data Agent:

A Fabric Data Agent is a Microsoft Fabric feature that turns your own data - lakehouses, warehouses, semantic models, KQL databases - into an AI agent that answers plain-language questions by generating the right queries behind the scenes, and can be shared or published into Microsoft 365 Copilot, Copilot Studio, or Azure AI Foundry.

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Assumption

1. The attendees has an existing Fabric Capacity Tenant (F2 and above)
2. Attendees must have a basic knowledge on how to navigate through Fabric Workspace
3. Pre-work has been completed by the IT admin of the client.
 - a. Setting up the Fabric Data Agent / AI and Ontology in Fabric Tenant Settings
 - b. Creation of folders for the attendees

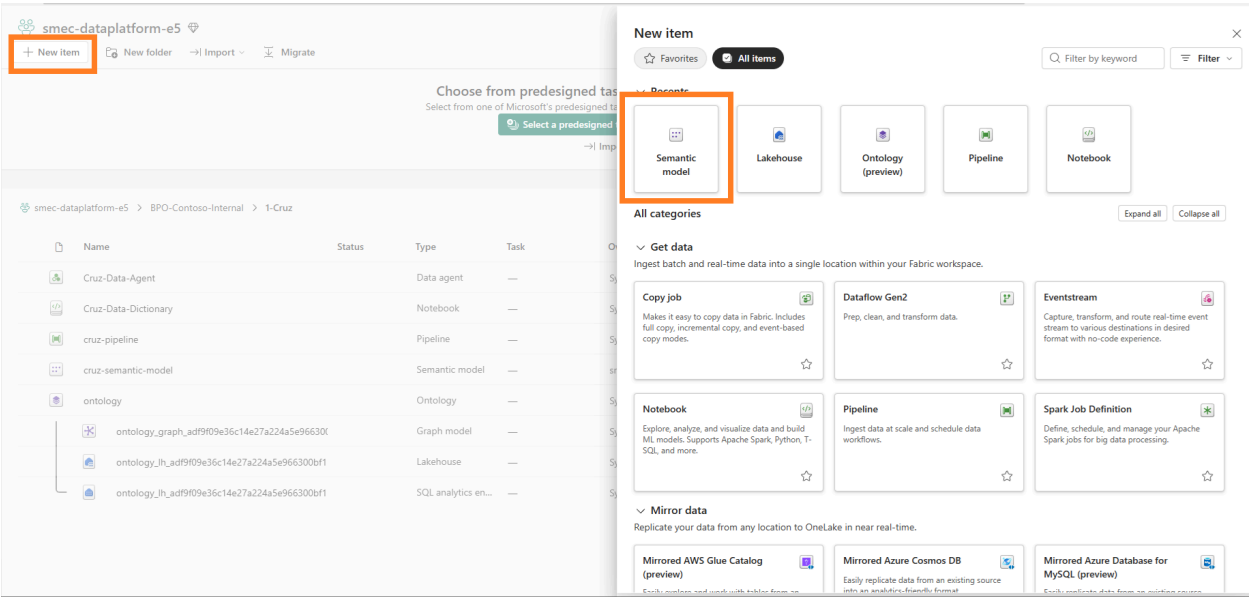
- c. Provide Fabric access to the attendees
- d. Lakehouse has been added together with the sample dataset

Let's Get Started!

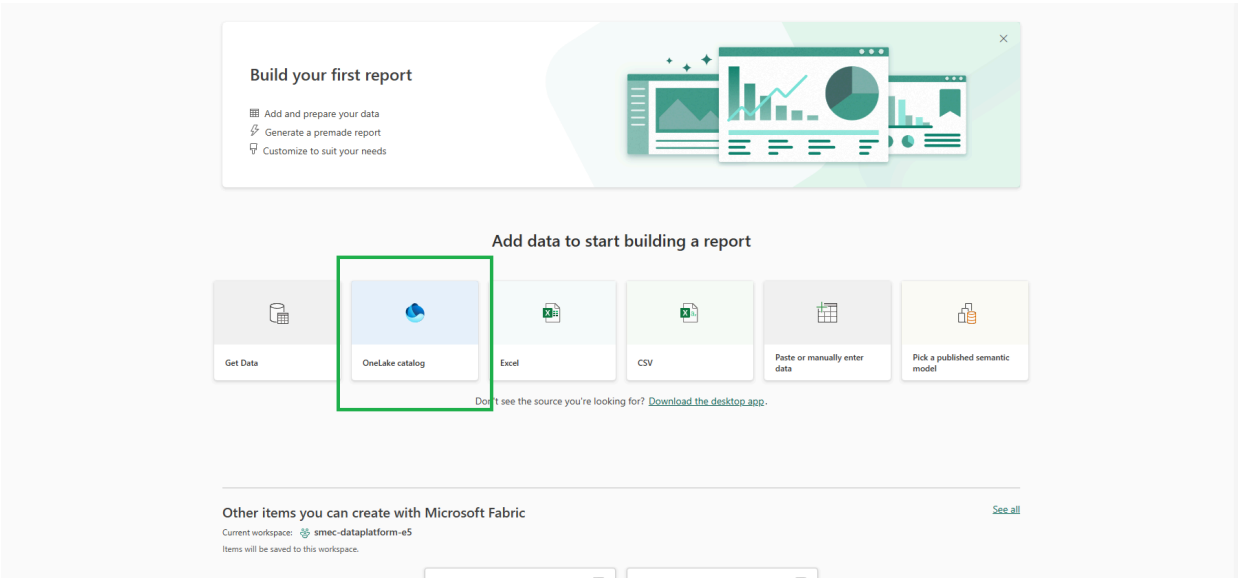
Configuring your Semantic Model

1. Create your own Semantic Model

- Create new item > select Semantic Model



- In the initial page, select One Lake Catalog



- Select the TasksUs Lakehouse

Discover data from across your org and use it to create reports

[All](#)
[Endorsed in your org](#)
[My data](#)
[Favorites](#)

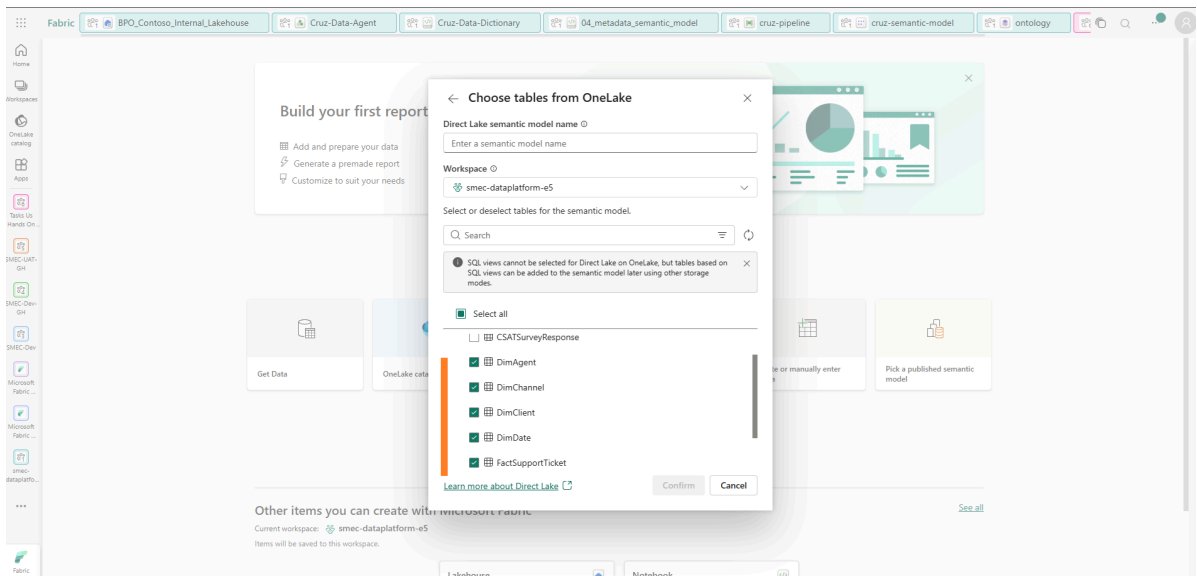
[Filter](#)

Name	Owner	Refreshed	Location	Endorsement	Sensitivity
TasksUs_Lakehouse	System Administrator	—	Tasks Us Hands On ...	—	—
BPO_Contoso_Internal_Lakehouse	System Administrator	—	smec-dataplatform-...	—	—
bpo_lakehouse_gh	System Administrator	—	SMEC-UAT-GH	—	—
agri_lakehouse	System Administrator	—	smec-dataplatform-...	—	—
bpo_lakehouse	System Administrator	—	SMEC-uat	—	—
bpo_lakehouse_gh	System Administrator	—	SMEC-Dev-GH	—	—
bpo_sql_database	System Administrator	—	SMEC-Dev-GH	—	—
bpo_lakehouse	System Administrator	—	SMEC-Dev	—	—
dataverse_v2smece5payg_cds2_workspace_u...	System Administrator	—	smec-dataplatform-...	—	—
mirrored-product-dataset	System Administrator	—	smec-dataplatform-...	—	—
Scan-capture-app	System Administrator	—	smec-dataplatform-...	—	—

[Connect](#)
[Cancel](#)

[Learn more about connection options](#)

- Select the following Tables
 - DimAgent, DimChannel, DimClient, DimDate, FactSupportTicket
 - DO NOT select CSATSurveyResponse

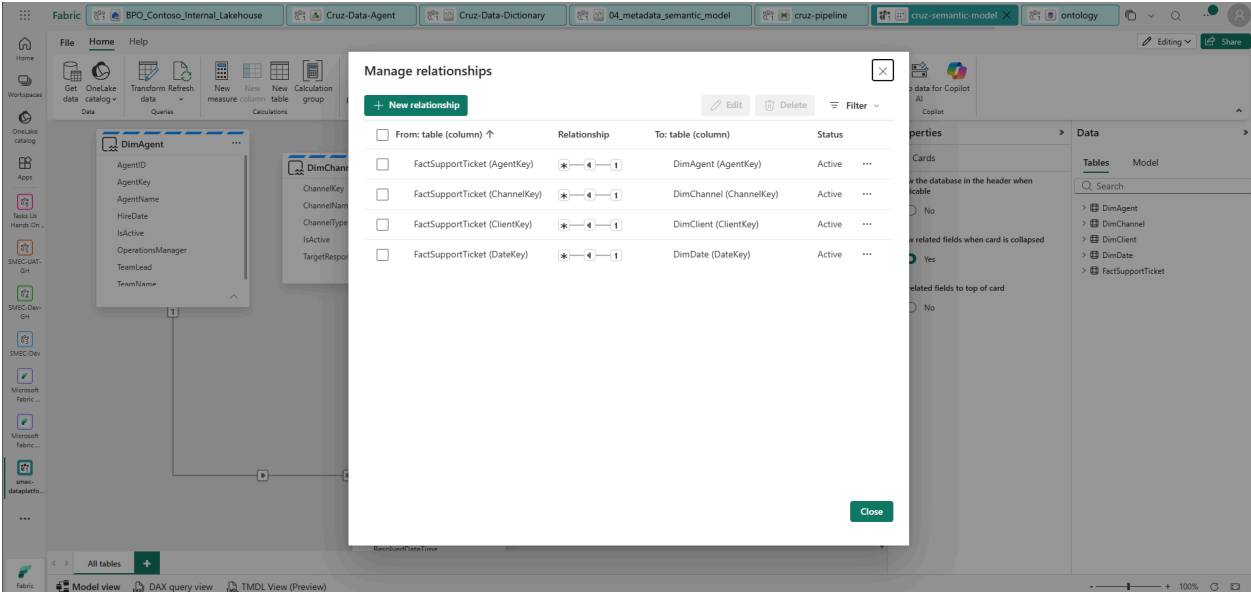


2. Onboarding Process #1 - Providing Business Context

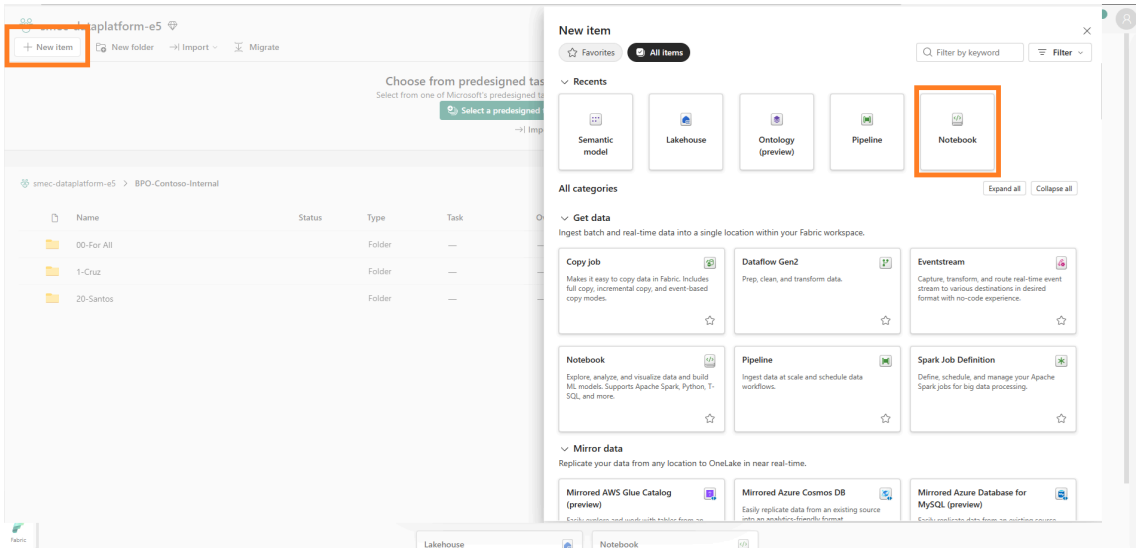
- Create relationship between tables
 - Click Your semantic model
 - Upper Ribbon click > Manage Relationships
 - Create New relationship
 - Make sure the relationship is 1:Many
 - Sometimes fabric will assume, but best if you can review and check

- Make sure the CSAT table is **not included** in your semantic model

From	To	Cardinality
FactSupportTicket.DateKey	DimDate.DateKey	Many-to-one
FactSupportTicket.AgentKey	DimAgent.AgentKey	Many-to-one
FactSupportTicket.ClientKey	DimClient.ClientKey	Many-to-one
FactSupportTicket.ChannelKey	DimChannel.ChannelKey	Many-to-one
CSATSurveyResponse.TicketKey	FactSupportTicket.TicketKey	One-to-one (optional)



- Provide Business Description into your semantic model
 - Reference [From Hours to Seconds: Automating Power BI Metadata in Fabric | by Therese Sario | Medium](#)
 - Step 1 - Create a Notebook
 - New Item > Select Notebook



- Open your newly created notebook, run the following python codes
 - Reference: [Fabric-Hands-On-BPO/02_notebook/Data Dictionary Notebook/Data Dictionary.ipynb](#) at main · sariotherese/Fabric-Hands-

Install necessary packages

```
%pip install semantic-link-labs
```

Code to autopopulate the descriptions

```
# Code to autopopulate
# All lines with ** - replace the values

import pandas as pd
from semy_labs.tom import connect_semantic_model

DATASET = "cruz-semantic-model" # as it appears in
the workspace **
WORKSPACE = "smec-dataplatform-e5" **
DESC_FILE = "Files/landing-zone/2_Data Dictionary.csv"
# lakehouse path, or use abfss://... **

# --- load the external file (CSV here; use pd.read_excel for .xlsx) ---
meta = pd.read_csv(f"/lakehouse/default/{DESC_FILE}")
meta["column_name"] = meta["column_name"].fillna("").astype(str).strip()
meta["table_name"] = meta["table_name"].astype(str).strip()
meta["description"] = meta["description"].fillna("").astype(str)

# index for quick lookup
tbl_desc = {r.table_name: r.description
             for r in meta.itertuples() if r.column_name
             == "" and r.description}
col_desc = {(r.table_name, r.column_name): r.description
            for r in meta.itertuples() if r.column_name
            and r.description}

applied, missing = 0, []

with connect_semantic_model(dataset=DATASET, workspace=
WORKSPACE, readonly=False) as tom:
    for t in tom.model.Tables:
        # table-level description
        if t.Name in tbl_desc:
```

```

        t.Description = tbl_desc[t.Name]
        applied += 1
    # column-level descriptions
    for c in t.Columns:
        key = (t.Name, c.Name)
        if key in col_desc:
            c.Description = col_desc[key]
            applied += 1

    # report anything in the file that didn't match a real object
    model_tables = {t.Name for t in tom.model.Tables}
    model_cols = {(t.Name, c.Name) for t in tom.model.Tables for c in t.Columns}
    missing = [k for k in tbl_desc if k not in model_tables]
    missing += [k for k in col_desc if k not in model_cols]

    print(f"Applied {applied} descriptions.")
    if missing:
        print("No matching object for:", missing)
    # Changes are committed automatically when the `with` block exits.

```

Expected Output

```

44 # Changes are committed automatically when the `with` block exits.
[9] ✓ 13 sec - Command executed in 7 sec 229 ms by System Administrator on 8/17/2026, 11:38:26 AM PySpark (Python)
> Log ...
...
Applied 46 descriptions.
No matching object for: [('CSATSurveyResponse', 'SurveyResponseKey'), ('CSATSurveyResponse', 'TicketKey'), ('CSATSurveyResponse', 'SurveyResponseDate'), ('CSATSurveyResponse', 'AgentCourtesyRating'), ('CSATSurveyResponse', 'ResolutionQualityRating'), ('CSATSurveyResponse', 'WaitTimeRating'), ('CSATSurveyResponse', 'FirstContactResolution'), ('CSATSurveyResponse', 'NPS_Score'), ('CSATSurveyResponse', 'CustomerComment'), ('CSATSurveyResponse', 'ResponseTimeHours')]

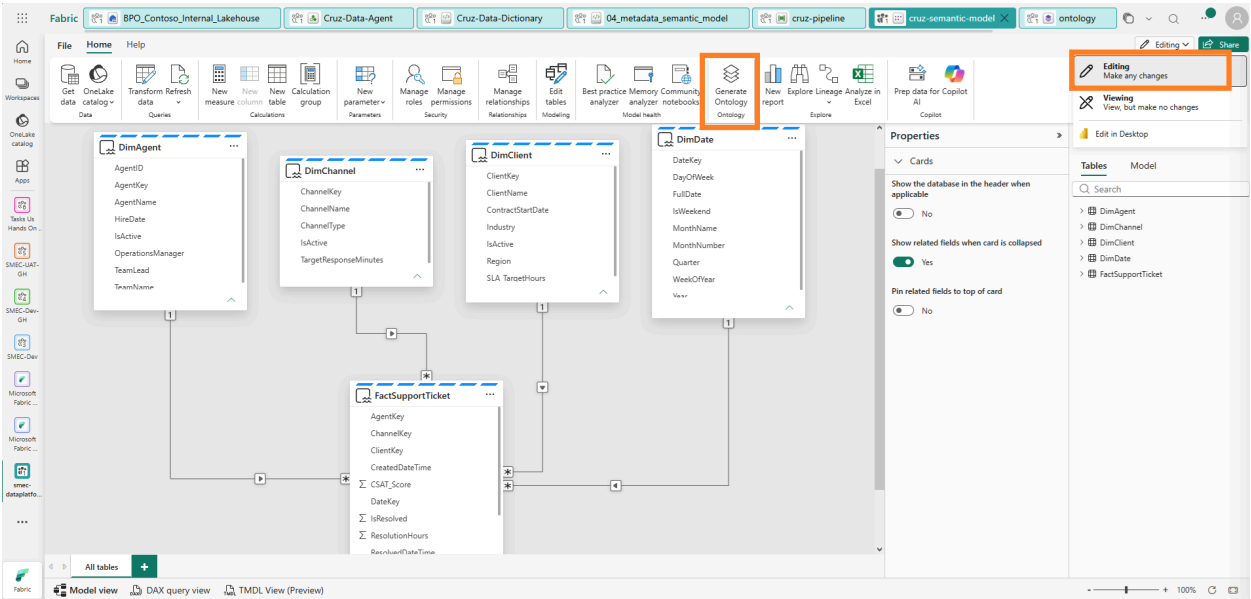
```

Creation of Ontology

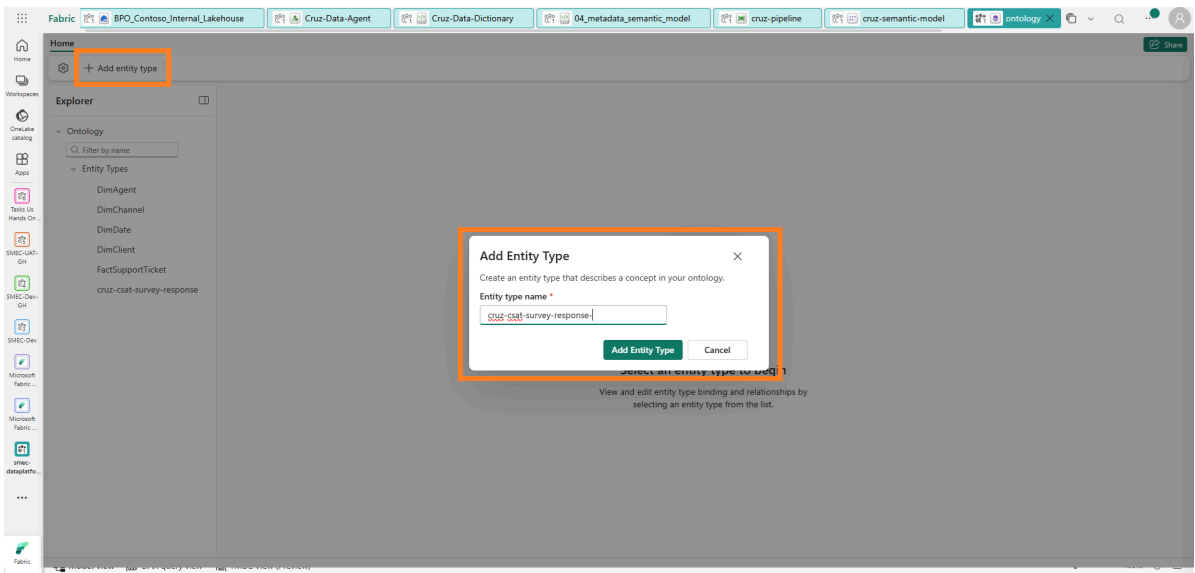
What is Ontology?

Fabric Ontology (preview, part of Fabric IQ) is a shared business-context layer that defines your enterprise concepts - entity types, properties, relationships - once and binds them to real data across OneLake, so people and AI agents can query and reason in business terms instead of tables and joins.

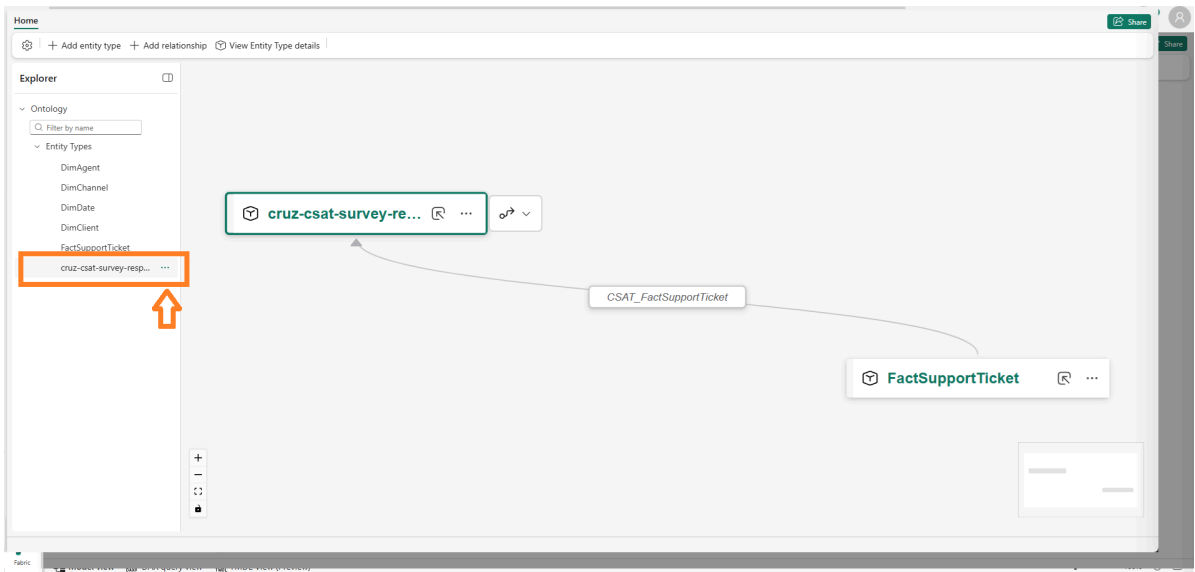
1. Go back to your semantic model
2. Upper right side > change from Viewing to Editing
3. Upper Ribbon click Generate Ontology
 - a. Rename the ontology: cruz-ontology



1. Go to your ontology
 - a. Click 'Add entity type'
 - b. rename entity : cruz-csat-survey-response

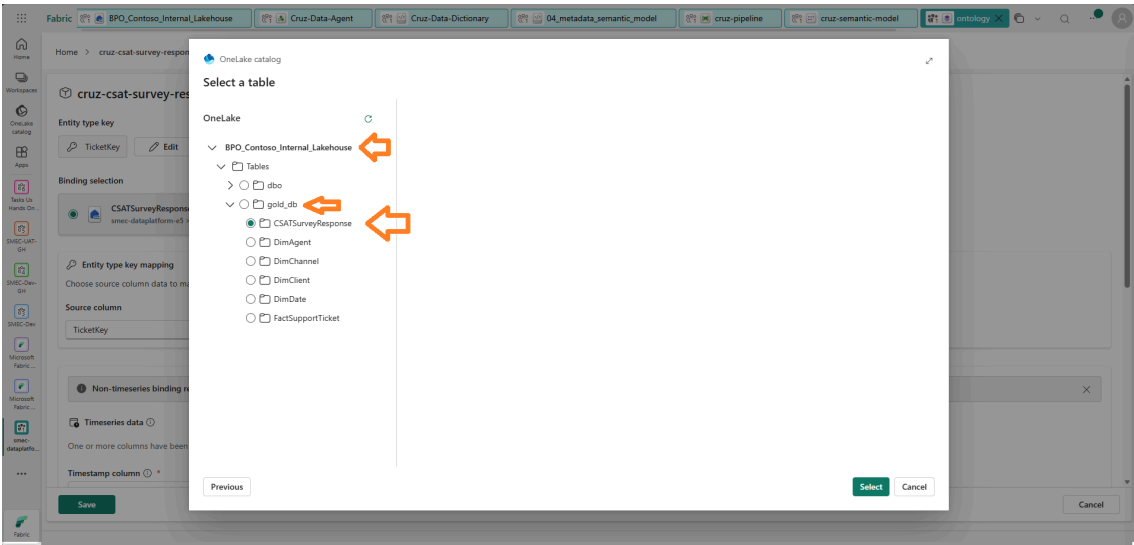


1. On the newly created entity, click the three dots > Select Bind data

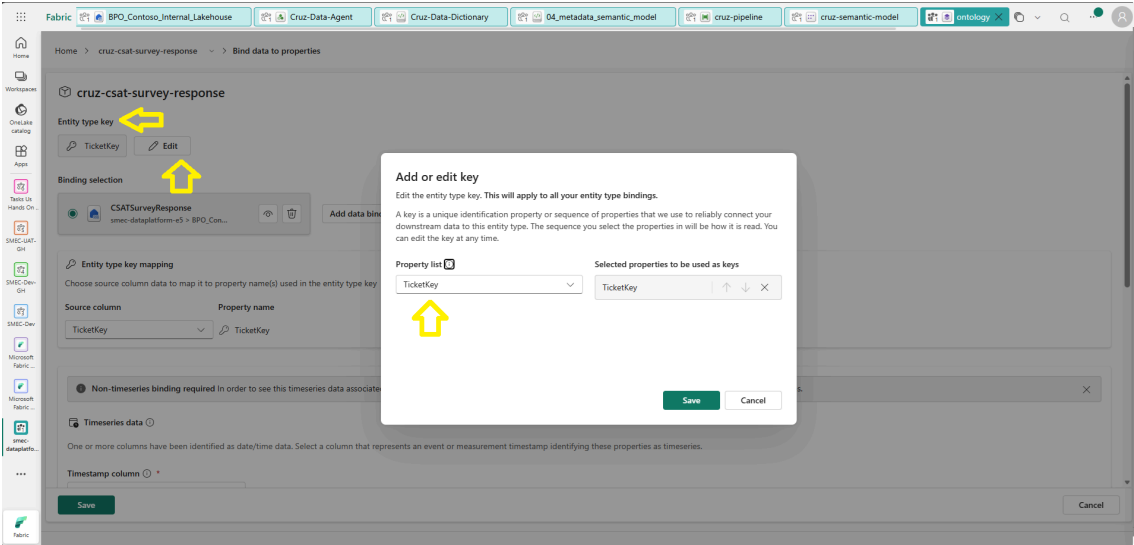


1. Under Binding Selection > Click Add Data binding > Lakehouse Table

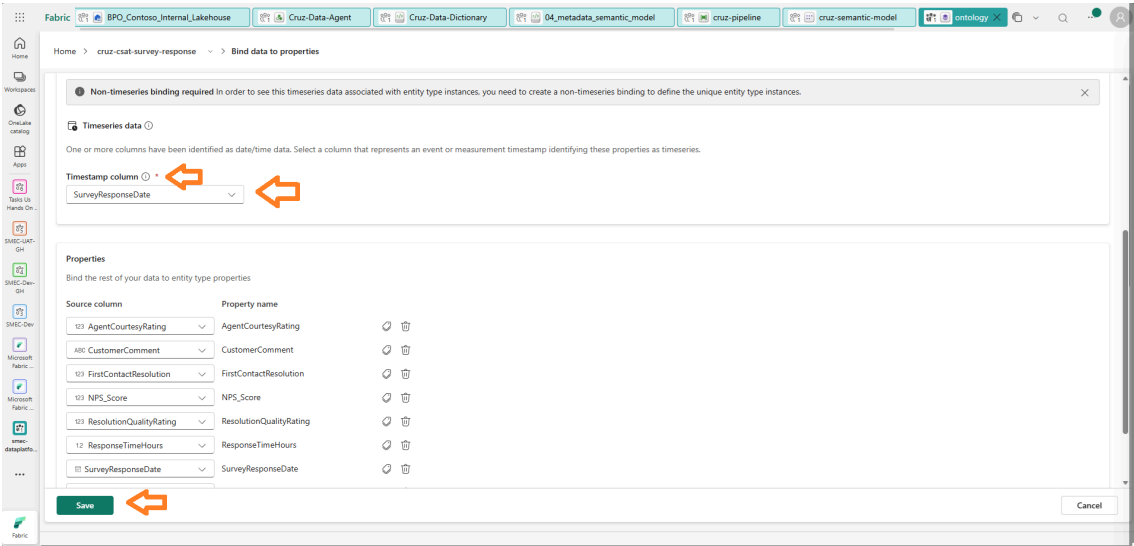
- a. Select your working lakehouse
- b. Select CSATSurveyReponse table only.



1. Under Entity Type key > Select: TicketKey



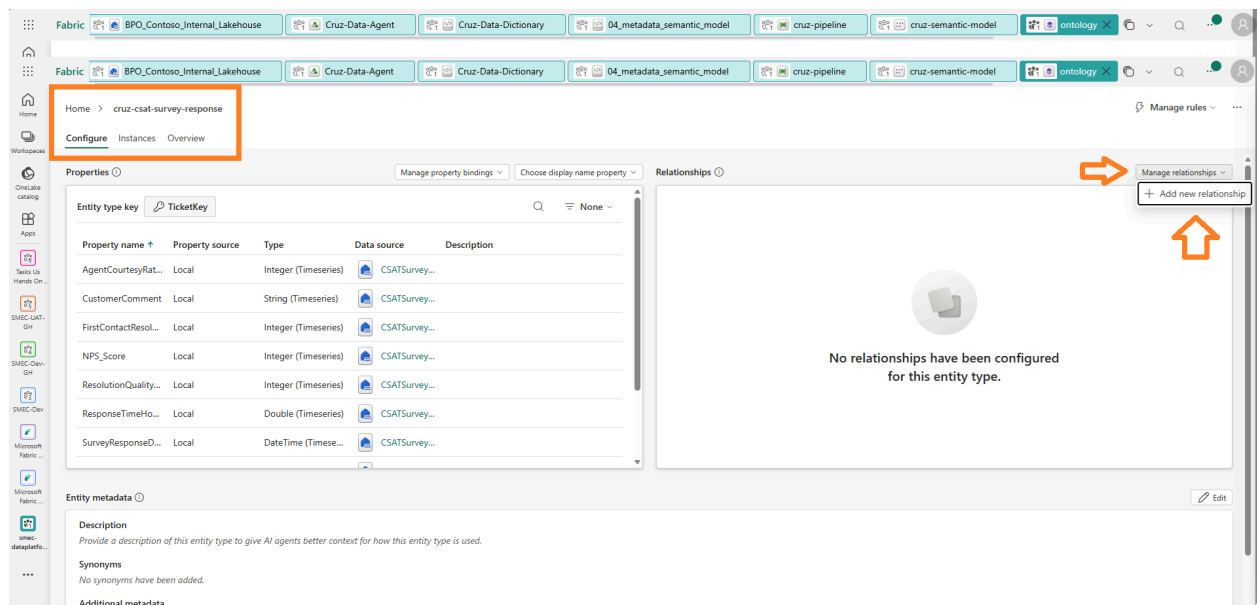
- 1. Under Entity Type key mapping > source column: TicketKey
- 2. Time Series Data > Timestamp column > SurveyResponseDate



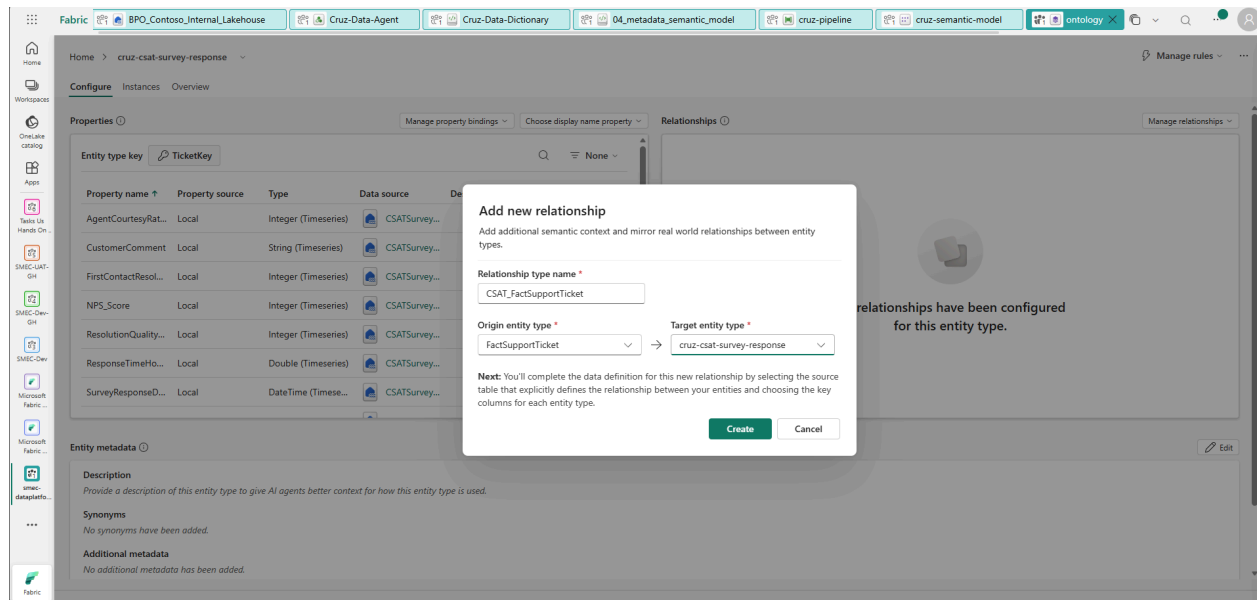
Click Save

2. Click Add Relationship

-

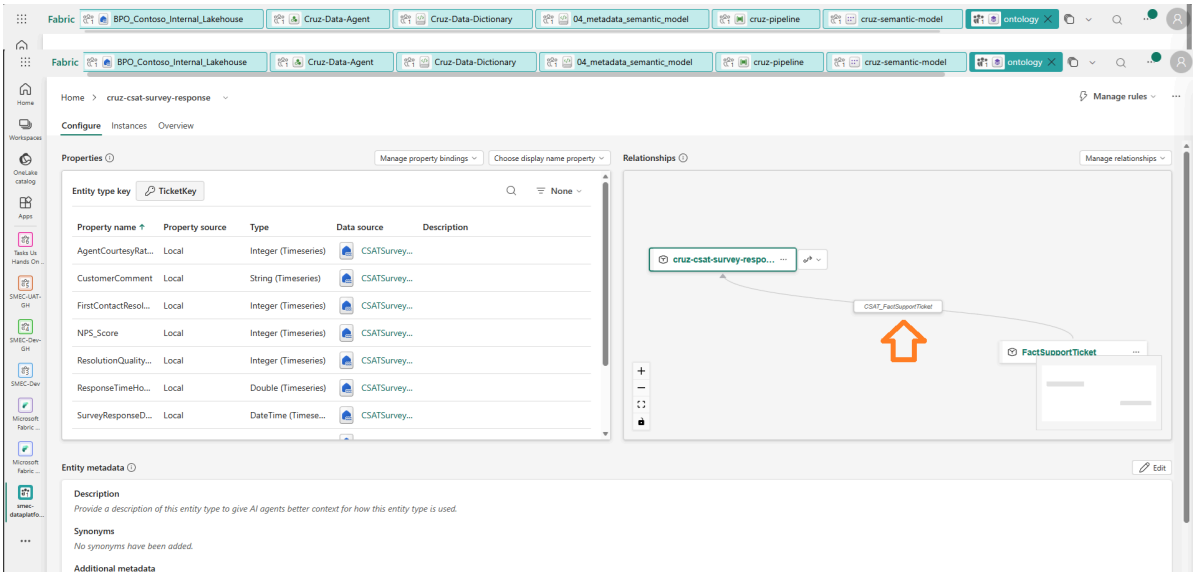


1. Relationship type name: CSAT_FactSupportTicket
2. Origin Entity Type: FactSupportTicket
3. Target Entity Type: cruz-csat-survery-response

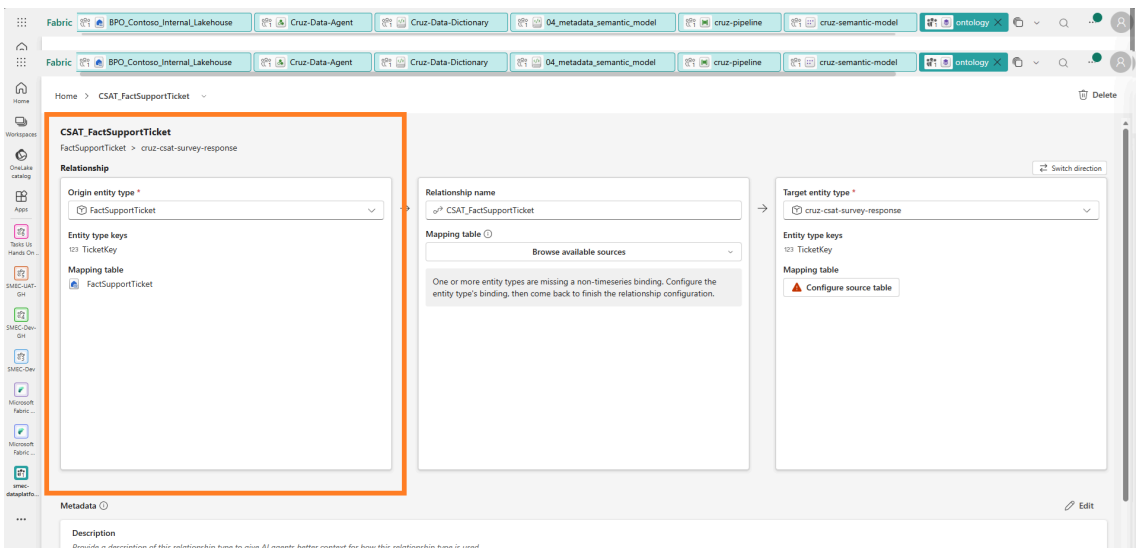


Click Create

1. For the Source Go to the Ontology Mapping
 - a. Click the CSAT_FactSupportTicket

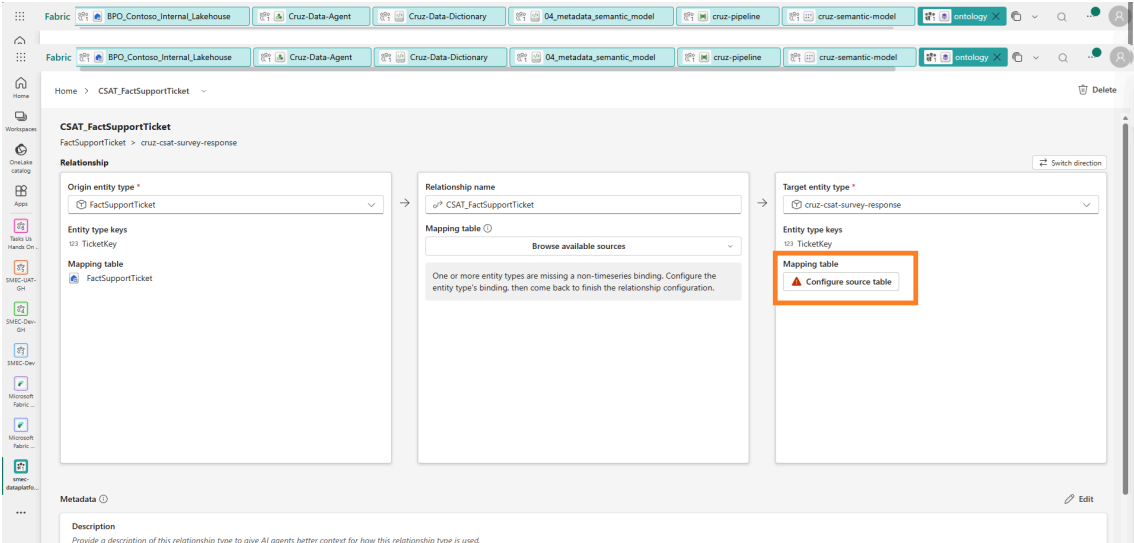


1. Under Relationship > Origin Entity Type: FactSupportTicket > Click the Configure Entity Type keys
 - a. Click Entity Type Key > Select in the drop down 'TicketKey'
 - b. Under Timeseries data > None

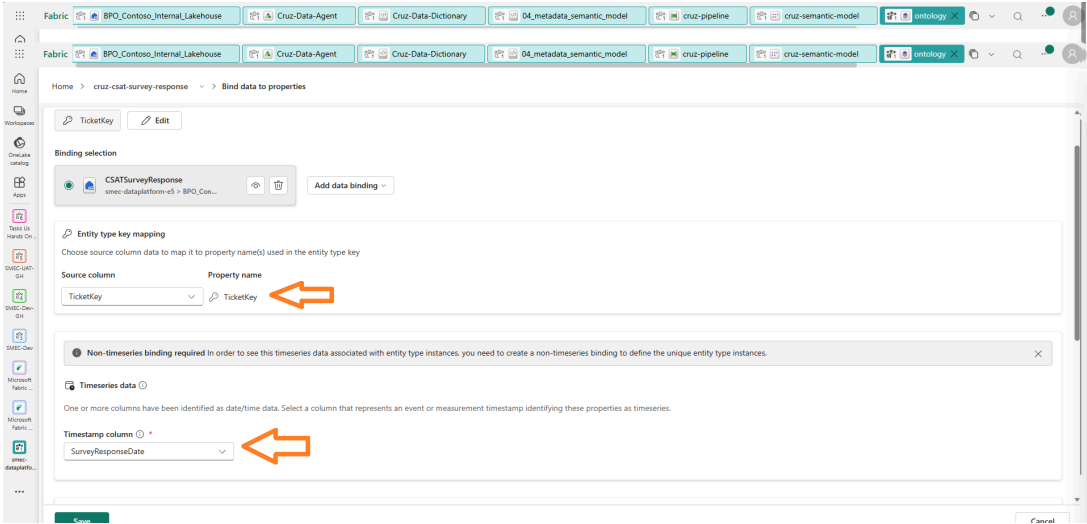


Click Save

2. For the Target: Go back to Ontology Mapping
 - a. Click the CSAT_FactSupportTicket
 - b. Under Relationship > Target Entity Type > Click Configure Source Table



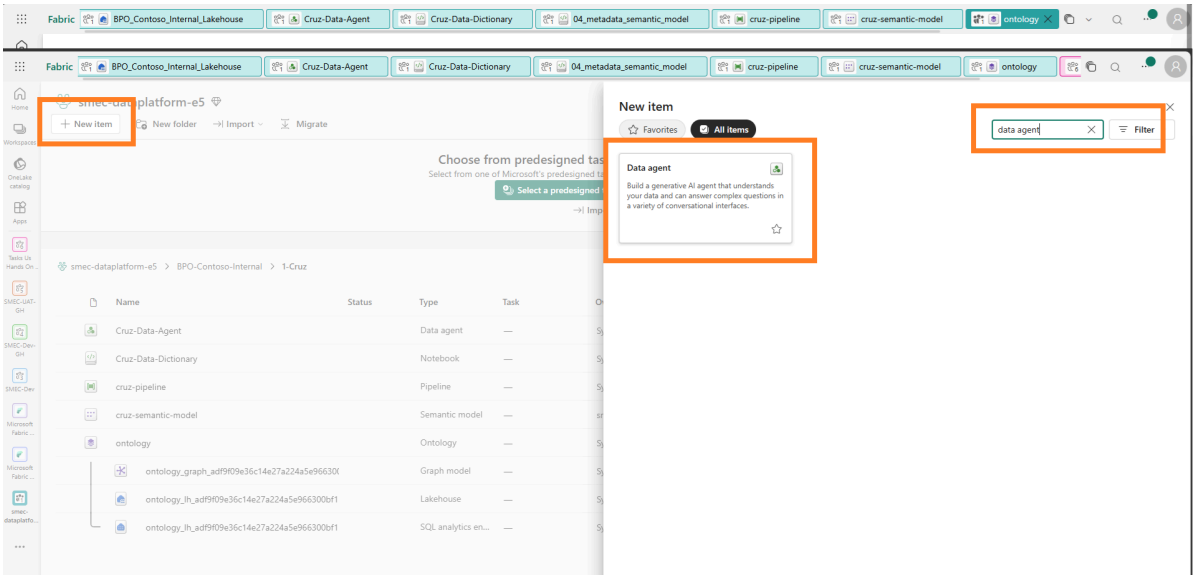
- 1. Make sure the Entity Type: TickeyKey
- 2. Timestamp Column: SurveyResponseDate



Click Save

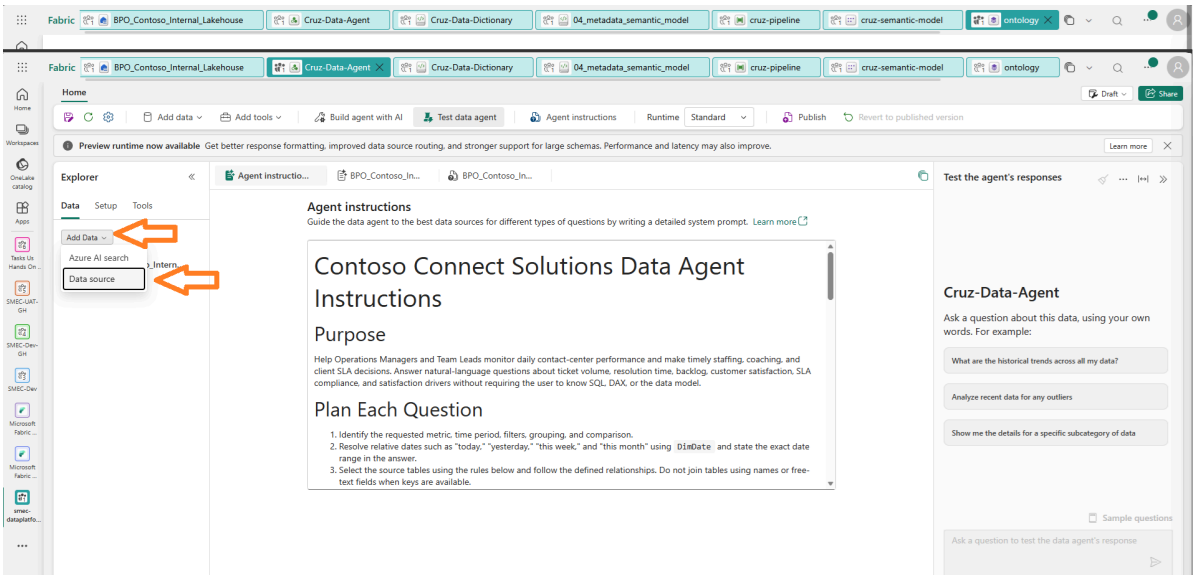
Creation of Fabric Data Agent

- 1. Go to your folder
- 2. Create new item >Data Agent
 - a. Name: <name>-Data-Agent



3. Add Data Items

a.

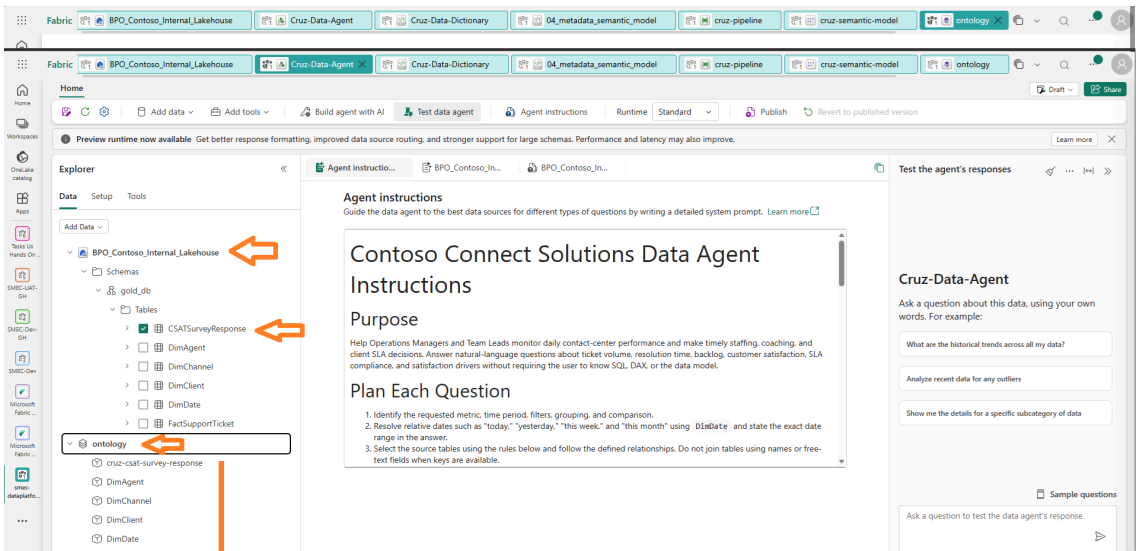


1. Select two data sources:

a. BPO_Contoso_Lakehouse

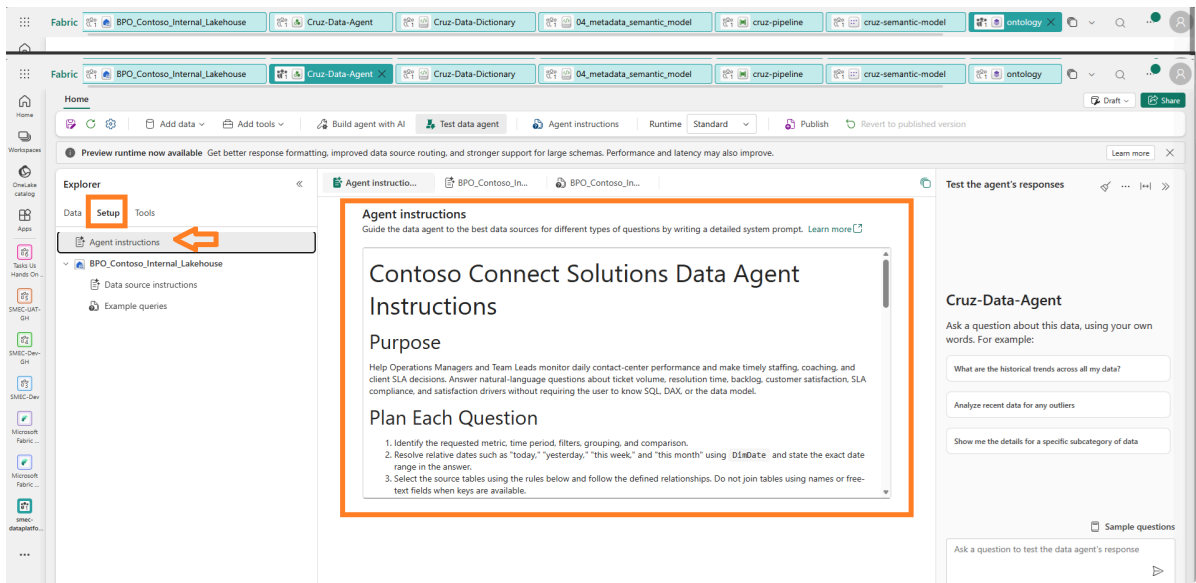
i. select only 'CSATSurveyResponse'

b. Ontology - auto select all

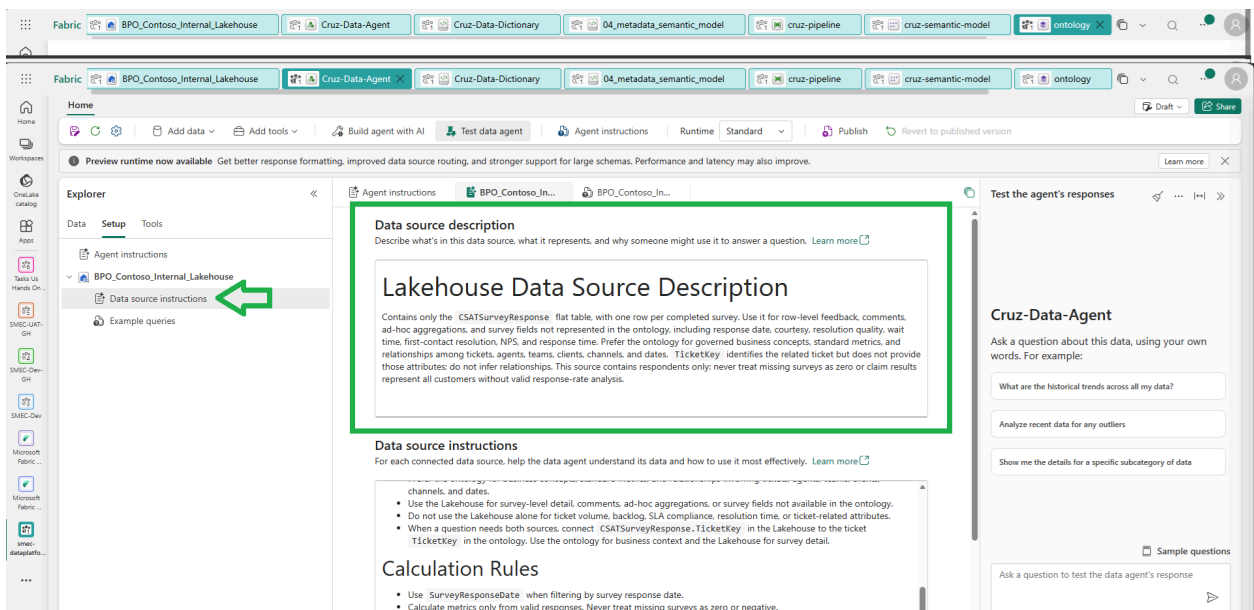


Onboard your Data Agent

- 1. Go to Setup Tab > Agent Instructions
 - a. Under the Agent Instructions, input the following
 - b. Go to Github, and input the following Agent Instructions: [Fabric-Hands-On-BPO/03_Fabric Data Agent/Agent Instruction at main · sariotherese/Fabric-Hands-On-BPO](#)

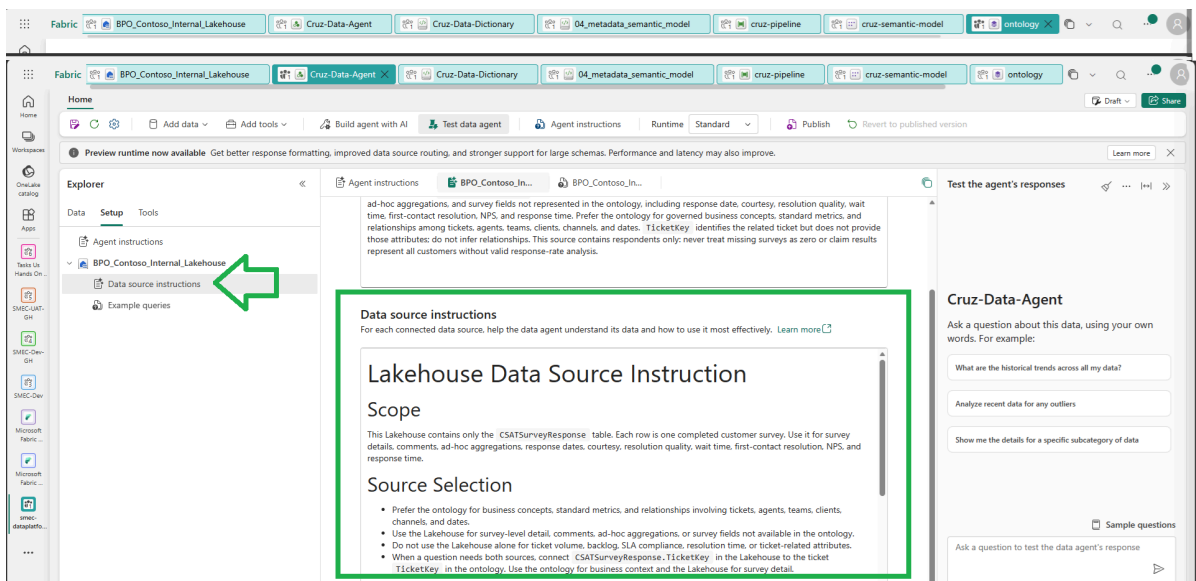


- 2. Go to Setup Tab > Data Source Instructions (For your Lakehouse)
 - a. Under the Data Source Description, paste the description,
 - b. Get from Github [Fabric-Hands-On-BPO/03_Fabric Data Agent/LH-Data Source Description.md at main · sariotherese/Fabric-Hands-On-BPO](#)



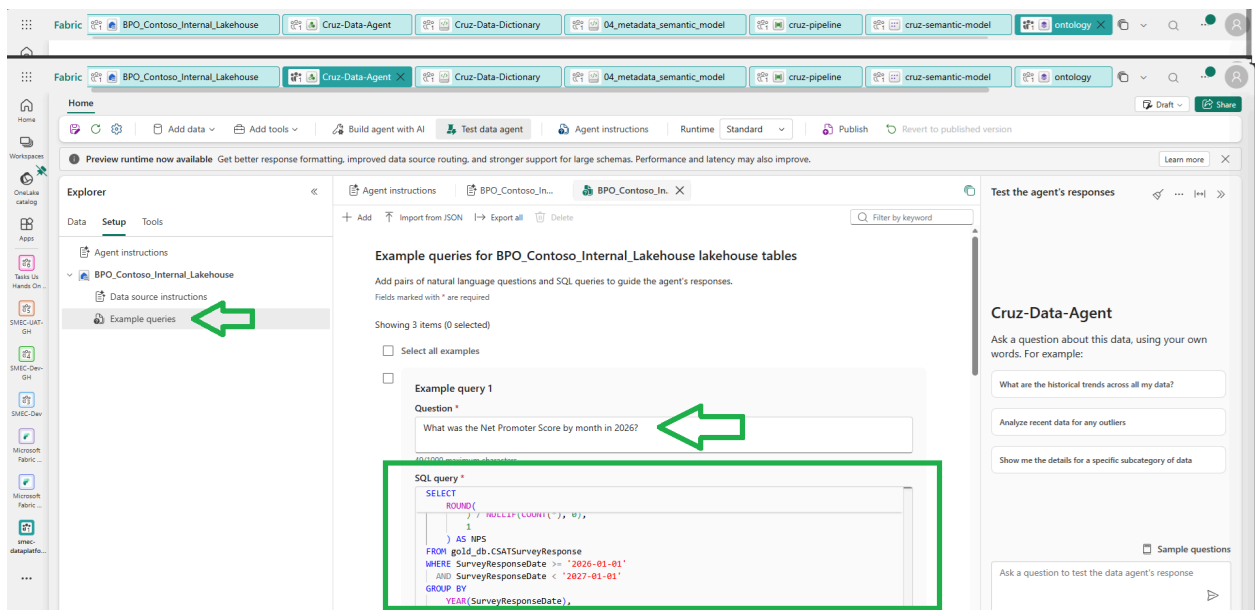
- 3. Still under Setup Tab > Data Source Instruction
 - a. Under Data Source Instructions
 - b. Populate using the Github values: [Fabric-Hands-On-BPO/03_Fabric Data Agent/LH-Data Source Instruction.md at main · sariotherese/Fabric-](#)

Hands-On-BPO



3. Sample Queries

- Populate the Sample Questions, and Sample Queries
- Use this Github Repo for reference: [Fabric-Hands-On-BPO/03_Fabric Data Agent/CSATSurveyResponse Business Questions.md at main · sariotherese/Fabric-Hands-On-BPO](#)



Test your Agents

Sample Questions for the Data Agent

- How many tickets were created from August 10-13, 2026, and how many are still open?
- Which team had the highest average resolution time this week?
- Show me the agents with the lowest average CSAT for Client A in July 2026.
- Show me the low-rated comments from Client A last January 2025?
- What is our NPS by channel?